TCG-Gel-15000: Thermal Gel

Deep Materials

Product Description

TCG-Gel-15000 is a thermally conductive single-part interface gel engineered to deliver effective heat dissipation across complex surface geometries and variable gap tolerances, while reducing stress transfer and maintaining reliable thermal contact in high-performance electronic systems. The silicone-based formulation combines high flow characteristics, low volatility, and electrically insulating properties to support stable long-term operation in reliability-critical applications.

Key Features

- High conductivity and low thermal impedance
- Stable performance at elevated temperatures
- Demonstrated thermal cycling stability
- Electrically insulating to minimize component cross-talk
- Compatible with automated dispensing systems
- RoHS compliant

Applications

- CPU, GPU & AI/HPC Thermal Interfaces
- SSD, DRAM & DIMM Cooling
- 5G RRU, RF & Telecom HW
- ADAS ECU, LiDAR & Radar Modules
- PSU, VRM & DC-DC Power Electronics
- BMS, IPC & Edge AI Systems

Product Naming

TCG-Gel-15000

- 15000 → Model: 15.0 W/m·K
- TCG-Gel → Single Part Thermal Gel

Application Guidelines

- Avoid exposure to open flame and direct sunlight.
- Ensure application surfaces are clean.
- Compatible with automated and manual dispensing systems.
- Avoid air entrapment during dispensing.

Storage and Shelf Life

- Store at 8°C to 28°C in a dry environment (RH<70%).
- Shelf life is 24 months from the date of manufacture.

Packaging Information

- Available in 30 cc (100 g) and 300 cc (1 kg) dual cartridge packaging
- Other sizes are available upon request

Property	Value	Test Method
Color	Gray	Visual
Flow Rate (g/min)	10	30cc tube, 0.090” nozzle, 90psi
Thermal Conductivity (W/m·K)	15.0	ASTM D5470
Minimum BLT (µm)	250	--
Density (g/cc)	3.5	ASTM D792
Breakdown Voltage (kV/mm)	>8.0	ASTM D149
Operating Temperature Range (°C)	- 50 to 200	--
Volatility (%)	0.13	ASTM E595
Flammability Rating	V-0	UL94
Volume Resistivity (Ω-cm)	$>10^{13}$	ASTM D257
Heat Capacity (J/g·K))	1.1	ASTM E1269

The data provided are typical values derived from laboratory testing and are intended for reference only. They should not be used as specifications or as a basis for design without independent validation.